

Preface

The mining industry is characterized by diverse geological, technological and operational conditions, with varying degrees of inherent hazards and associated risks. In this context, the adoption of a **risk-based approach at the mine level** has assumed increasing importance for prioritizing resources, identifying critical areas of concern and implementing preventive measures for reducing the likelihood and severity of mine accidents.

With a view to strengthening this approach at the national level, it is considered desirable to develop a robust and transparent framework for assessing the **risk profile, regulatory compliance and overall safety performance of individual mines**. Such a framework would facilitate data-driven prioritization of regulatory interventions, encourage continuous improvement in safety performance and promote a culture of self-regulation and proactive risk management within the mining industry.

Accordingly, a Committee was constituted to examine and formulate a framework for the **National Risk Rating Index (NRRI)** and the **National Compliance Rating Index (NCRI)**. The Committee has submitted its report containing recommendations regarding the methodology, parameters, indicators and scoring framework for assessment of risk and compliance at the national level.

The proposed integration of the **NRRI** and **NCRI** would provide the basis for determining an **Overall Mine Safety Rating (OMSR)**, which is intended to provide a holistic assessment of a mine by taking into account both its inherent risk profile and its safety management and regulatory compliance performance. The proposed framework is envisaged as an instrument for **prioritizing inspections and regulatory resources, identifying areas requiring focused intervention, encouraging improved safety performance and recognizing mines demonstrating exemplary safety standards**.

The proposed rating framework is also intended to facilitate benchmarking of mine safety performance against recognized good practices and, where appropriate, internationally accepted approaches, thereby promoting **continuous improvement, responsible mining and self-regulation**.

Before finalization of the methodology and scoring framework for the **NRRI, NCRI and OMSR**, it is considered appropriate to obtain wider views and expert inputs from all concerned stakeholders. Accordingly, the **report of the Committee is being placed in the public domain for comments and suggestions** from mine operators, workers' representatives, researchers, academicians, professional and technical institutions, industry experts and other stakeholders associated with mine safety and allied fields.

Stakeholders are invited to critically examine the proposed parameters, indicators, weightages, scoring methodology and overall framework and furnish **specific, evidence-based and constructive comments/suggestions within fifteen (15) days** from the date of publication of this document. The inputs received will be examined appropriately and may be considered while finalizing the **National Risk Rating Index, National Compliance Rating Index and Overall Mine Safety Rating framework**.

The objective of this consultation is to develop a **credible, transparent, objective, practicable and continuously improvable national mine safety rating system** that strengthens preventive safety management, facilitates effective regulatory intervention and promotes a sustained culture of **risk management, compliance and self-regulation** across the Indian mining industry.

CHAPTER I

BACKGROUND

1.0 Background Information:

- 1.1 Mining, by its very nature, is a high-risk economic activity involving complex interactions between geological, technical, and human factors. In India, where mining contributes significantly to the national economy and supports critical sectors such as energy, infrastructure, and manufacturing, the scale of operations is vast ranging from deep underground coal mines and large opencast metal mines to extensive oil and gas fields. Given its inherent hazards and socio-economic importance, the need for a robust, scientific, and proactive safety management framework is paramount. The Indian mining-safety regulatory framework is governed by the Occupational Safety, Health and Working Conditions Code, 2020, along with the rules, regulations, and orders made thereunder, and is administered by the Directorate General of Mines Safety (DGMS). Over time, the regulatory framework has gradually evolved from a prescriptive, rule-based approach towards a risk-based and performance-oriented model of safety governance. In this evolving regulatory environment, two complementary indices the National Risk Rating Index (NRRI) and the National Compliance Rating Index (NCRI) have been conceptualized to modernize the assessment of mine safety and promote sustainable and responsible mining practices.
- 1.2 Building upon these frameworks, the integration of the National Risk Rating Index (NRRI) and the National Compliance Rating Index (NCRI) leads to the development of an Overall Mine Safety Rating (OMSR), which holistically represents both the inherent risk profile and the safety management performance of a mine. This composite rating enables a balanced evaluation - recognizing that even a high-risk mine can operate safely if robust compliance systems, monitoring technologies, and proactive safety cultures are in place. Conversely, low-risk mines with poor compliance can still pose significant hazards. By combining these two indices through a weighted or normalized model, the OMSR provides a scientific and transparent basis for classifying mines, prioritizing inspections, allocating regulatory resources, and incentivizing good safety practices. In the Indian regulatory context, such an integrated rating system supports DGMS in data-driven decision-making, continuous safety improvement, and sustainable resource development aligned with national safety and ESG (Environmental, Social, and Governance) goals.
- 1.3 Earlier National Safety Award (NSA) was conferred to mines annually based on LAFP (Longest Accident-Free Period), LIFR (Lowest Injury Frequency Rate), LIFRM (Lowest Injury Frequency Rate on Man shift), LIFRO (Lowest Injury Frequency Rate based on output (composite of ore/coal and overburden)). The criteria adopted was not comprehensive.

2.0 Committee

- 2.1 Mandate of the Committee was to suggest methodologies for determining new National Index for standardization of Risk Rating & Compliance Rating mines. This calculated index will be based on a measurable Risk Index and Compliance.

Index shall be prepared to each categorization of mines (Coal/Metal/Oil), which will incorporate the existing list of the mines, statutory compliance by respective mine managements. The index will incorporate all statutory compliance parameters, safety statistics and other measurable indicators.

2.2 Proceedings of the Committee:

The committee held its meeting virtually on 05.08.2025, 08.10.2025 and 09.10.2025. Meeting was chaired by Sri Niraj Kumar, DDG, NZ, Ghaziabad. The details of the members attending the meeting are enclosed as Annexure-1.

Committee discussed in detail the following:

There is no system to categorize mines on national level based on of the inherent risks and compliance and devise intervention strategy accordingly. The National Risk Rating Index (NRRI) is a composite framework developed to assess, compare, and monitor the safety and risk profile of mines across India.

It provides a quantitative risk rating to each mine based on multiple parameters related to operational, geological, environmental, and human factors.

The main idea is to:

- Identify mines prone to accidents or disasters.
- Prioritize safety interventions.
- Promote a proactive, risk-based approach instead of a reactive one.
- The concept aligns with the "Vision Zero" Accident Policy of the Government of India and aligned with the objective of Directorate General of Mines Safety.

CHAPTER II

CATEGORIZATION & SUB-CATEGORIZATION OF MINES

3.0 Categorization & sub-categorization of mines:

Committee discussed that for the purpose of risk and compliance rating it is required to categorize mines into coal, metalliferous and oil mines. Further, it was discussed that all coal, metalliferous and oil mines are also required to be further sub-categorized based on production and man shift so that there is a fair rating and compliance indexing within this group. The idea was that mines having different scale of operation will have different risk levels and different level of compliance requirement.

Accordingly, the annual return data of the mines submitted on DGMS Portal has been analyzed and following categorization has been finalized:

3.1 Metalliferous Mines:

Type	Small	Medium	Large
Underground	≤0.5 MT	>0.5 to ≤1.0 MT	>1.0 MT
Open Cast	≤0.5 MT	>0.5 to ≤5.0 MT	>5.0 MT

MT is million ton; MCM is million cubic meters

Note: In case of open cast mines, production figure includes total mineral and overburden handled in a calendar year.

3.2 Coal Mines:

Type	Small	Medium	Large	Mega Projects
Underground	≤0.5 MT	>0.5 to ≤1.0 MT	>1.0 to ≤2.0 MT	>2.0 MT
Open Cast	≤0.5 MCM	>5.0 to ≤20.0 MCM	>20.0 to ≤40.0 MCM	>40.0 MCM

3.3 Oil Mines:

There will be four categories of mines in oil sector in which two categories in Production Mines, namely Small and Large based on annual production in MMSC (Million Metric Standard Cubic Meters) and the other two will be Workover Mine and Drilling Mine. Further where any mine consists of all these three or two operations, the mine may be classified based on production and the risk rating of each operation shall calculated and be clubbed to get the weighted score of the mine.

Production Mines		Work Over Mines	Drilling Mines
Small	Large		
≤ 7 MMSC	> 7 MMSC		

3.4 **Criteria for Mixed (both Opencast & Underground) mines:**

The annual production would be taken as prime evaluating factors for the mines of mixed category. In cases, where the production in opencast mines exceeds that of underground mines, they should be considered as opencast mines, and vice versa. For mixed mines, the overall score shall be normalized based on the risks calculated separately for the underground and open-cast operations.

CHAPTER III

NATIONAL RISK RATING SYSTEM

The following major risk factors are considered for Coal, Metalliferous and Oil Mines separately. Further, in coal & metalliferous mines risk factor has been separately considered for belowground and opencast mines.

Risk Rating system has been proposed in three formats, covering (i) belowground mines (both coal and metalliferous), (ii) Open cast (both coal and metalliferous), and (iii) Oil mines. Risk rating system assigns risk score for broad categories based on risk score generated on the basis of risk values assign for risk sub-categories. The assignment of risk score shall be system based, on the basis of the data submitted by mine management.

Total score on adding the individual category score gives the inherent Risk score of a mine. Risk score arrived for each risk category is residual risk score which is difference of inherent risk score and risk reduction score. Risk reduction score is calculated on the basis of mitigatory measures taken by mine management against that particular risk category or sub-category.

Accordingly, format for submission of details by individual mine has been designed to consider all risk categories, mitigatory measures and a manner to arrive at value of residual risk. This format is designed separately for (i) belowground mines (both coal and metalliferous), (ii) Open cast (both coal and metalliferous), and (iii) Oil mines.

Format as mentioned above has been attached as part of Annexure to this report.

4.0 RISK SCORING SYSTEM:

4.1 Belowground (Coal & Metalliferous) Mines:

4.1.1 Geological and Geotechnical Risks (Total category score: 200)

- Roof and side conditions (**sub-category score: 50**)
- Rock Mass Rating or any other technical classification ratings (50)
- Presence of geological disturbances (50)
- Presence of water-bearing strata or weak beds (50)

4.1.2 Operational Risks (1000)

- Mining methods including multi-seam working (100)
- Mechanization (50)
- Depth (50)
- Equipment condition and maintenance (50)

- Man-Machine interaction (50)
- Production Level (50)
- Operational area (large, medium and small) (50)
- Existence of old workings (50)
- Explosives and blasting practices (50)
- Ventilation (50)
- Dust control (50)
- Illumination (50)
- Presence of inflammable gases (100)
- Fire and Spontaneous heating (50)
- Inrush of water from underground sources (100)
- Inrush of water from surface sources (50)
- Electricity (50)
- Danger from adjoining mines (50)

4.1.3 Supervision (100)

- Adequacy of supervision (200)
 - Manager (50)
 - Assistant Managers (30)
 - Safety Officers (10)
 - Engineers (20)
 - Surveyors (10)
 - Mining Supervisor (20)
 - Assistant Mining Supervisors (20)
 - Mechanical Supervisors (20)
 - Electrical Supervisors (20)

4.1.4 Human and Management Factors (150)

- Safety training (50)
- Safety management system & audits (50)
- safety & compliance culture (50)

4.1.5 Emergency Preparedness (200)

- Availability of rescue facilities (100)

- Emergency Response Plan (50)
- Disaster management plans (25)
- Communication and monitoring systems (25)

4.1.6 History (100)

- History of Accidents (50)
- History of Dangerous Occurrences (50)

4.2 Opencast Mines (Coal & Metalliferous) Mines:

4.2.1 Geological and Geotechnical Risks (300)

- Pit slope stability (100)
- Dump slope stability (100)
- Presence of geological disturbances (50)
- Presence of water-bearing strata or weak beds (50)

4.2.2 Operational Risks

- Mining method and depth (100)
- Equipment condition and maintenance (50)
- Man-Machine interaction (50)
- Maintenance of Haul Road (50)
- Maintenance of LMV Road (50)
- Explosives and blasting practices (50)
- Dust control (50)
- Illumination (100)
- Fire (50)
- Inrush of water (50)
- Electricity (50)

4.2.3 Supervision (100)

- Adequacy of supervision (200)
 - Manager (50)
 - Assistant Managers (30)
 - Safety Officers (10)
 - Engineers (20)
 - Surveyors (10)

- Mining Supervisor (20)
- Assistant Mining Supervisors (20)
- Mechanical Supervisors (20)
- Electrical Supervisors (20)

4.2.4 Human and Management Factors (150)

- Safety training (50)
- Safety management system & audits (50)
- safety and compliance culture (50)

4.2.5 Emergency Preparedness (100)

- Emergency Response Plan (50)
- Disaster management plans (25)
- Communication and monitoring systems (25)

4.2.6 History (100)

- History of Accidents (50)
- History of Dangerous Occurrences (50)

4.3 Oil Mines:

4.3.1 Operational Risks (750)

- Equipment condition and maintenance (100)
- Man-Machine interaction (100)
- Explosives and Fracturing practices (50)
- Dust control (50)
- Illumination (50)
- Fire (100)
- Explosion (100)
- Blowout (100)

4.3.2 Supervision (100)

- Adequacy of supervision (220)
 - Agent (20)
 - Manager (50)
 - Assistant Managers (30)
 - Safety Officers (30)

- Dy. Manager (20)
- Engineers (20)
- Fire Officer (20)
- Competent person for well perforation (10)
- Mechanical / Electrical Supervisors (20)

4.3.3 **Human and Management Factors (150)**

- Safety training (50)
- Safety management system & audits (50)
- safety and compliance culture (50)

4.3.4 **Emergency Preparedness (100)**

- Emergency Response Plan (50)
- Disaster management plans (25)
- Communication and monitoring systems (25)

4.3.5 **History (100)**

- History of Accidents (50)
- History of Dangerous Occurrences (50)

Format as mentioned above has been attached as part of Annexure to this report as furnished below:

Coal Mines (belowground)	Annexure I
Metalliferous Mines (belowground)	Annexure II
Coal Mines (opencast)	Annexure III
Metalliferous Mines (opencast)	Annexure IV
Oil Mines	Annexure V

CHAPTER IV

NATIONAL RISK RATING INDEX

5.1 Classification of Mines as per Normalized Residual Risk Score

Based on the computed residual risk score in the previous chapter, the normalized risk score on a scale of 100 will be calculated. Each mine will be classified on the basis of this normalized risk score into three categories, high risk mine, medium risk mine, low risk mine. Indicative action like frequency of inspection and mitigatory measure required has been suggested on the basis of category in which a particular mine falls.

The category of mine is as below:

Category	Normalized Residual Risk Score	Risk Level	Indicative Action
Category A	More than 70	High Risk Mine	Intensive inspection and monitoring; immediate mitigation measures required
Category B	>30 - ≤70	Medium Risk Mine	Regular inspection; risk reduction plan implementation
Category C	Below 30	Low Risk Mine	Routine safety supervision; maintain good practices

Note: Above risk score is normalized risk score on a scale of 100 by adding all individual risk scores and scaling to normalization.

5.2 National Risk Rating Index:

All mines of the country will be rated category wise on the basis of the normalized risk score of that particular mine. A national level risk based profiling of mines will be created on the basis of this.

5.3 Importance of the National Risk Rating Index

- (i) Scientific Risk Management, provides a quantitative, data-driven approach to mine safety management rather than relying on subjective judgement.
- (ii) Prioritization of Inspections, helps DGMS and mining companies to allocate resources and inspections based on actual risk rather than size or output alone.
- (iii) Reduction of Accidents and Disasters, by identifying weak areas early, preventive actions can be prioritized based on category or sub category wise residual risk score.
- (iv) Benchmarking and Performance Evaluation, allows comparison between mines and encourages competition for safety excellence.
- (v) Public Accountability and Transparency, the index can be made publicly available to promote responsible mining.

CHAPTER V

NATIONAL COMPLIANCE RATING INDEX

- 6.1 The National Compliance Rating Index (NCRI) is an assessment system designed to evaluate how well mines comply with safety laws, standards, and best practices as prescribed in the legislation and regulatory bodies.

Unlike the Risk Rating Index, which measures how risky a mine's environment is, the Compliance Index measures how effectively a mine's management and workforce follow safety systems, rules, and preventive measures. It provides a quantitative score for each mine, reflecting its overall safety culture, management effectiveness, and regulatory compliance.

The NCRI framework will evaluate following major dimensions of mine safety compliance, each consisting of several measurable parameters:

6.1.1 Regulatory Compliance

Adherence to the OSH&WC Code, 2020 (erstwhile the Mines Act, 1952)

- The Coal Mines Regulations, 2017/ Metalliferous Mines Regulations, 1961 / Oil Mines Regulations, 2017 (250)
- The OSH&WC (Central) Rules, 2026 (200)
- The Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 (50)
- Timely submission of statutory reports & returns (25)
- Notice issued under Sec. 38(1)B(a) and Sec. 38(1)B(k) of the OSH&WC Code, 2020 (50)
- Order issued under Sec. 38(1)B(b), 38(1)B(c), 38(1)B(d) and Sec. 38(1)B(l) of the OSH&WC Code, 2020 and in force for more than a year (50)
- Long pending Violations (more than six months) (50)

6.1.2 Safety Management System (SMS) Implementation

Safety Management Plan & its implementation (100)

Framing & implementation of Standard Operating Procedures (SOPs) (50)

Safety awareness campaigns, mock drills, and toolbox talks (25)

Involvement of trade unions and workers in safety activities (25)

6.1.3 Action post-accident & dangerous occurrences

Reporting of accidents & dangerous occurrences (25)

Analysis & follow up action after accidents & dangerous occurrences. (100)

Near-miss reporting and analysis (25)

6.1.4 Training, Awareness, and Worker Participation

Compliance of provisions related to the OSH&WC (Central) Rules, 2026 (200), Regular refresher and vocational training programs (100)

6.1.5 Technology and Monitoring Systems

Use of modern safety technologies (e.g., proximity sensors, slope stability radar, real-time gas monitoring etc.) (100)

Maintenance of safety equipment and PPE standards (50)

Accordingly, format for submission of details by individual mine has been designed to consider all compliance categories and a manner to arrive at value of compliance score. This format is designed separately for (i) belowground mines (both coal and metalliferous), (ii) Open cast (both coal and metalliferous), and (iii) Oil mines.

Format as mentioned above has been attached as part of Annexure to this report as furnished below:

Coal Mines (belowground)	Annexure VI
Metalliferous Mines (belowground)	Annexure VII
Coal Mines (opencast)	Annexure VIII
Metalliferous Mines (opencast)	Annexure IX
Oil Mines	Annexure X

6.2 Methodology for Arriving at Compliance Score:

Each parameter receives a score weighted based on importance. The composite score determines the Mine's Safety Compliance Score.

Based on the computed compliance score as detailed above, the normalized compliance score on a scale of 100 will be calculated. Each mine will be classified on the basis of this normalized compliance score into four categories, Excellent, Good, Moderate and Poor. Indicative action like frequency of inspection and mitigatory measure required has been suggested on the basis of category in which a particular mine falls.

Total score on adding the individual category score gives the inherent Risk score of a mine. Risk score arrived for each risk category is residual risk score which is

difference of inherent risk score and risk reduction score. Risk reduction score is calculated on the basis of mitigatory measures taken by mine management against that particular risk category or sub-category.

6.3 Typical Classification of Mines Based on NCRI

Category	Compliance Rating	Safety Status	Indicative Action
Category A	85 – 100	Excellent	Model mine, showcase best practices
Category B	70 – 84	Good	Continue improvement, periodic review
Category C	50 – 69	Moderate	Immediate corrective action needed
Category D	< 50	Poor	Intensive DGMS monitoring, possible suspension of operations

6.4 Importance of the National Compliance Rating Index

- (a). Promotes a shift from compliance by enforcement to safety by commitment.
- (b). Provides an objective, measurable system for comparing safety performance between different mines, companies, and regions.
- (c). Regulatory Prioritization, helps DGMS to identify non-compliant mines quickly, Plan targeted inspections and audits, Allocate resources efficiently.
- (d). Incentive-Based System, High-rated mines can receive recognition, awards, or relaxed inspection frequency, while poor performers face stricter scrutiny or penalties.
- (e). Data-Driven Decision Making, Creates a national database of mine safety compliance, useful for Policy formulation, Research, Safety performance trend analysis, Accident prevention strategy.

A high NCRI score can reflect good governance and safety standards

CHAPTER VI

OVERALL MINE SAFETY RATING (OMSR)

- 7.1 **Overall Mine Safety Rating (OMSR)** should combine both NRRI and NCRI, showing how effectively a mine with a certain inherent risk level is managing its safety.

Conceptual Relationship

The relationship can be visualized as:

$$\text{"Safety Performance"} = f(\text{"Risk Level"}, \text{"Compliance Level"})$$

A high-risk mine with high compliance can still be safe.

A low-risk mine with poor compliance can become unsafe.

Hence, both indices must be balanced to represent true safety status.

7.2 Proposed Mathematical Model

NRRI: Normalized Risk Rating Index (0 to 100)

NCRI: Normalized Compliance Rating Index (0 to 100)

OMSR: Overall Mine Safety Rating (0 to 100)

Since higher NRRI means higher risk and higher NCRI means better safety, we need a formula where risk acts inversely to compliance.

Option 1: Weighted Inverse Relationship

$$OMSR = (W_1 \times NCRI) + [W_2 \times (100 - NRRI)]$$

Where:

W_1 = Weight for compliance (e.g., 0.6)

W_2 = Weight for risk mitigation (e.g., 0.4)

Option 2: Normalized Index (Balanced View)

$$OMSR = \frac{NCRI}{(NRRI+1)} \times C$$

where C is a scaling constant (≈ 100) to fit the output range 0 - 100.

This is simple, effective, and easy for annual benchmarking.

*After deliberations it was decided that Option 2, which gives **Normalized Index** is most suitable for Calculation of Overall Mine Safety Rating.*

7.3 Classification of OMSR (Overall Mine Safety Rating):

OMSR (%)	Category	Safety Status	Interpretation
85–100	A	Excellent	Well-controlled, model mine
70–84	B	Good	Safe operation, moderate improvement needed
50–69	C	Average	Some safety gaps, targeted improvements needed
35–49	D	Poor	High-risk and/or poor compliance, unsafe conditions
Less than 35	E	Very poor	Very High-risk and non-compliant, dangerous conditions

7.4 Advantages of the Combined Rating (Overall Mine Safety Rating):

Integrates both inherent hazards and human/system performance; Enables objective comparison across different mines types; Supports prioritized inspections and resource allocation; Encourages continuous improvement and competition for safety excellence; Provides a scientific tool for policy, ESG reporting, and investment decisions.

The establishment of unified indices such as the NRRI and NCRI, and their integration into an Overall Mine Safety Rating (OMSR), marks a decisive step toward a modern, data-centric safety governance system in India’s mining industry.

This framework not only enables objective comparison and benchmarking across diverse mine types—coal, metal, and oil—but also aligns regulatory oversight with the principles of risk-based inspection and preventive safety management.

By leveraging these indices, DGMS and mining organizations can jointly pursue continuous risk reduction, technology-driven monitoring, and capacity building at all operational levels.

CHAPTER VII

SUBMISSION PROCESS

8.1 Self-Submission forms:

Based on the framework, self-submission forms will be developed for different category of mines which will be filled by the mine management themselves digitally. The form will capture data objectively as far as practical to make the evaluation system more transparent and objective.

An Online portal shall be developed to facilitate submission of forms by the mine management. Mines which fail to digitally self-submit the prescribed forms within the stipulated time may either be assigned the lowest rating category, such as "E" rating, or an alternate mechanism may be developed for obtaining and entering the requisite information through empanelled third-party auditors or other authorized agencies at the cost of the mine management.

8.2 Verification System:

Online verification: The data submitted by the mine management in the online portal shall be verified with existing records and information available

Physical verification: The entries of the high overall safety rated mines shall be verified through physical inspections. Similarly, entries of the mine with abnormally low safety score may also be verified physically.

To ensure reliability and integrity of the submitted data, provision may also be made for periodic verification, random audits, cross-validation with inspection findings, and third-party assessment wherever considered necessary

This submission of the prescribed forms for the National Risk Rating Index (NRRI) and National Compliance Rating Index (NCRI) will be mandatory for all mines across the country, irrespective of category, mineral, size, or ownership.

Repeated non-submission or deliberate suppression of information may also be considered as an adverse compliance factor while determining the overall mine safety rating.

8.3 Software Development:

A dedicated and integrated software system may be developed for implementation of the National Risk Rating Index (NRRI) and National Compliance Rating Index (NCRI) framework. The software shall function as a centralized digital platform for submission, processing, validation, analysis, and monitoring of mine-related safety and compliance data across the country.

The software architecture may be designed to support scalability, interoperability, reliability, and secure handling of large volumes of data generated from mines of different categories and minerals. The system may include separate user interfaces for mine managements, DGMS officials, third-party auditors, and administrators with role-based access controls and authentication mechanisms.

The platform may provide online self-submission modules with predefined formats, automated data validation checks, field verification, logical consistency checks, and document upload facilities to minimize errors and improve quality of submitted data. Provision may also be incorporated for digital signatures, electronic acknowledgements, and audit trails to ensure authenticity and traceability of submissions.

The system should automatically generate National Risk Rating Index (NRRI), National Compliance Rating Index (NCRI), and Overall Mine Safety Ratings based on predefined algorithms.

Provision may further be made for integration of the software with existing government digital platforms such as the Shram Suvidha Portal to generate risk based inspection.

The software may also incorporate Artificial Intelligence (AI) and data analytics tools for identification of abnormal trends, predictive risk assessment, early warning generation, and prioritization of mines for inspections under the risk-based randomized inspection system.

At a later stage this system can be further upgraded for automated alerts for delayed submissions, adverse safety indicators, recurring violations, increasing accident trends, or sudden deterioration in compliance levels.

CHAPTER VIII

OUTCOME & RECOMMENDATIONS

9.1 Outcomes:

- (a). The proposed framework is expected to promote a culture continuous improvement and responsible mining practices.
- (b). The Overall Mine Safety Rating derived from the aforesaid indices is expected to promote accountability and healthy competition among mine operators towards achieving higher standards of safety and compliance.
- (c). The system may also incentivize timely reporting of accidents, dangerous occurrences, and near-miss incidents, thereby strengthening preventive safety culture in the mining sector.

9.2 Recommendations:

- (a). It is recommended that submission of the prescribed forms for the National Risk Rating Index (NRRI) and National Compliance Rating Index (NCRI) be made mandatory for all mines across the country, irrespective of category, mineral, size, or ownership.
- (b). The Overall Mine Safety Rating shall be publicized in web portal for information. This will serve as motivation for higher overall mine safety rating.
- (c). The overall mine safety rating shall be incorporated in Shram Suvidha portal and shall be factored into the risk-based randomized inspection selection process. Mines with Higher OMSR may be subjected to reduced inspection frequency. Mines with poor ratings may receive enhanced and targeted inspections.
- (d). Periodic review and updating of the indices may be undertaken to ensure alignment with current trends.
- (e). It is also recommended that a structured reward and recognition system be instituted for mines with higher Overall Safety Rating Index. As system of incentivising such mines by other government departments may also be thought upon. Such mines may be facilitated for grant of future leases and other clearances.
- (f). A detailed notification in this regard may be issued so as to formalise the whole process and give a legitimate dimension to it.